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CO 2 :- Second law of Thermodynamics

Limitations of first law of Thermodynamics :-

1. It does not explain about direction of heat flow b/w system and surroundings

2. It does not explain whether the process is spontaneous (or) non-spontaneous

Ex:- Melting of ice into water at room temperature is spontaneous.

All man-made reactions are non-spontaneous.

3. Reverse process is not possible

Ex:- Water to ice at room temperature

4. Practically it is not possible to convert complete heat energy into complete work ($Q \neq W$)

5. During the conversion of heat energy to work there is some portion of energy that is unavailable.

This is not explained in first law but explained in second law by introducing the topic Entropy.

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Second law of Thermodynamics :-
It is also called as law of Entropy
The change in entropy is equal to
amount of heat transfer / Temp in kelvin (K°)

Mathematically,

$$\Delta S = \frac{Q}{T} \left(\frac{J}{K^\circ}\right)$$

differentiatal form :-

$$ds = \frac{dQ}{T}$$

Concept of Entropy :- (S)
En - Inside
tro - converting unavailable energy.

1. It is unavailable form of energy that cannot be converted into work b/w system and surroundings (Universe)
2. It is the measurment of degree of disorder
3. It measures the randomness of system
4. It measures uncertainity of system, degree of energy dispersal

⇒ The entropy of universe always increases that means

$$\Delta S_{universe} \geq 0$$

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surrounding} \geq 0$$

⇒ The entropy of reversible process = 0

$$\Delta S_{\text{Reversible Process}} = 0$$

$\therefore \Delta S_{system} + \Delta S_{surrounding} = 0$

$\therefore \Delta S_{system} = -\Delta S_{surrounding}$

Tutorial :-

Prove that $C_p - C_v = R$ for ideal gas (4M)

C_p - Specific heat at constant pressure

C_v - Specific heat at constant volume

R - Gas constant

Specific heat of water	$= 4200 \frac{J}{kg \cdot C^\circ}$
Specific heat of ice	$= 3090 \frac{J}{kg \cdot C^\circ}$
Specific heat of water vapour	$= 3010 \frac{J}{kg \cdot C^\circ}$
Specific heat of human	$= 350 \frac{J}{kg \cdot C^\circ}$

Specific Heat :- The amount of heat energy required to increase 1°C per unit mass of the substance.

It can be measured at const. pressure and const. volume

Acc. to first law :-

$$\Delta U = Q - W$$

in differential form

$$dU = dQ - dW \quad \text{— (1)}$$

The change in internal energy (du) can be estimated from $C_p \cdot dT$. Similarly, the amount of heat energy transfered can be estimated $C_v \cdot dT$

for ideal gas:-

$$PV = RT \quad (dw = PV)$$

Sub in eq ①

$$C_v \cdot dt = C_p \cdot dT - d(PV)$$

$$C_v \cdot dT = C_p \cdot dT - RdT \quad (dPV = RdT)$$

$$\boxed{C_v - C_p = R}$$

Change in Entropy for ideal gas

22/01/26 ALM - 1 (CO-2)

A. Vasantha
2500010068

Prove that $C_p - C_v = R$ for ideal gas

C_p - Specific heat at const. pressure
C_v - Specific heat at const. volume
R - Gas constant

Acc. to first law of thermodynamics:-

$$\Delta U = Q - W$$

$$dU = dQ - dW \quad \text{— ①}$$

Q - heat
W - work
U - Internal energy.

The change in dU can be $C_p \cdot dT$. Heat energy transfered is $C_v \cdot dT$

for ideal gas

$$PV = RT \quad (dW = PV)$$

Sub in eq ①

$$C_v \cdot dT = C_p \cdot dT - d(PV)$$

$$C_v \cdot dT = C_p \cdot dT - RdT \quad (dPV = RdT)$$

$$\boxed{C_v - C_p = R}$$

22/1/26.

Change in Entropy for ideal gas

Acc. to first law

$$\Delta U = Q - W$$

in differential form

$$du = dQ - dw$$

$$\therefore dQ = du + dw \quad \text{—— (1)}$$

Acc. to second law

$$ds = \frac{dQ}{T} \quad \text{—— (2)}$$

then sub eq (1) in eq (2) and we get

$$ds = \frac{du + dw}{T}$$

then further;

$$ds = \frac{du}{T} + \frac{dw}{T} \quad \text{—— (3)}$$

where change in internal energy can be estimated through specific heat at const. volume and the differential change in temperature.

In mathematical way;

$$du = C_v \cdot dT \quad \text{—— (4a)}$$

for 1M of ideal gas :- $PV = RT$

$$P = \frac{RT}{V}$$

where,

$$dw = d(pV) = pdV$$

$$= \left(\frac{RT}{V}\right) dV \quad \text{—— (4b)}$$

Sub (4a), (4b) in (3)

$$ds = \left(\frac{C_v \cdot dT}{T}\right) + \frac{R\cancel{T}}{\cancel{V}} \frac{dV}{\cancel{T}} \quad \text{—— (5)}$$

Integrating on both sides

$$\int ds = C_v \int_{T_1}^{T_2} \frac{dT}{T} + R \int_{V_1}^{V_2} \frac{dV}{V}$$

$$\Delta S = C_v (\log T)_{T_1}^{T_2} + R (\log V)_{V_1}^{V_2}$$

$$\boxed{\Delta S = C_v \log\left(\frac{T_2}{T_1}\right) + R \log\left(\frac{V_2}{V_1}\right)} \quad \text{— (6)}$$

Case 1 :- Const. Temperature :-

$(T_1 = T_2)$ $\therefore \log\left(\frac{T_2}{T_1}\right) = \log T = 0$

$$\therefore \Delta S = C_v(0) + R \log\left(\frac{V_2}{V_1}\right)$$

$$\boxed{\Delta S = R \log\left(\frac{V_2}{V_1}\right)} \quad \text{— (7)}$$

Case 2 :- Const. volume.

$(V_1 = V_2)$ $\therefore \log\left(\frac{V_2}{V_1}\right) = \log v = 0$

Sub in eq (6)

$$\Delta S = C_v \log\left(\frac{T_2}{T_1}\right) + R(0)$$

$$\boxed{\Delta S = C_v \log\left(\frac{T_2}{T_1}\right)} \quad \text{— (8)}$$

Eqn (6), (7), (8) are used to estimate the entropy change for ideal gas.

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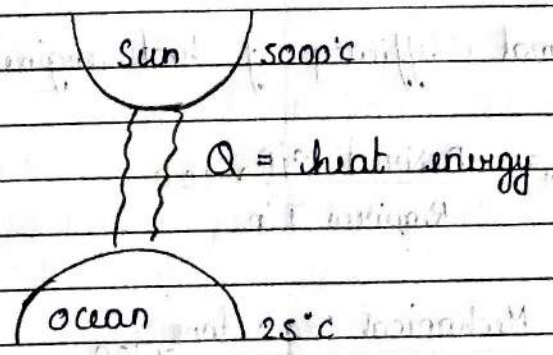
Applications of laws of Thermodynamics

Thermal Reservoirs :-

These are thermodynamic systems consists of heat source and heat sink

Heat Source	Heat Sink
It provides large amount of heat.	It absorbs large amount of heat with negligible change in temperature
It maintains high temperature	Maintains at low temperature
Ex:- Sun, human body, boiler, Geothermal rocks, Internal-Combustion	Ex:- Ocean bodies, lakes, rivers, surroundings, condensors in industry

Heat transfer from sun to ocean bodies

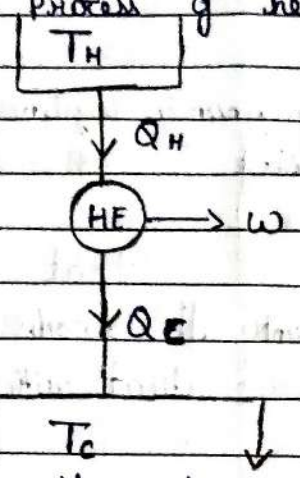


Applications for Heat Transfer :-

Laws of thermodynamics are used to solve the practical problems related to heat transfer in process industries such as 1. Heat Engine
2. Refrigerator
3. Heat pump.

Heat Engine (HE) :-

The Process of heat engine diagram :-



It is a thermodynamic device which converts heat energy to mechanical work.

It consists hot reservoir at a temperature T_H (°K) and cold reservoir at a temperature T_C.

During conversion heat energy (Q_H) into mechanical work (ω) some portion of energy is rejected to the cold reservoir (Q_C)

The Thermal efficiency of heat engine can be defined as

$$(\eta)_{HE} = \frac{\text{Desired O/P}}{\text{Required I/P}} \times 100$$

$$= \frac{\text{Mechanical Work done}}{\text{Heat energy Transfer}} \times 100$$

Mathematically :-

$$\boxed{\eta_{HE} = \frac{\omega}{Q_H} \times 100} \quad \text{—(1)}$$

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As per law of conservation of energy (Ist law)
Energy balance of heat engine can be written as

Heat IN = Heat OUT
(System) (Surrounding)

$Q_H = W + Q_C$

$\therefore \boxed{W = Q_H - Q_C}$ —— ②

Sub eqn ① in eqn ② we can get

$$\eta_{HE} = \left(\frac{Q_H - Q_C}{Q_H}\right) \times 100$$

A 2,200 = 8
B - 3,600
= 800

$$= \boxed{\left(1 - \frac{Q_C}{Q_H}\right) \times 100} \quad \text{—— ③}$$

$Q_H = \frac{116}{28}$

Q_C and Q_H are unmeasurable quantities and can be estimated through their corresponding temperatures

$Q_H \propto T_H \Rightarrow Q_H = KT_H$

$Q_C \propto T_C \qquad Q_C = KT_C$

$$\therefore \eta_{H.E} = \left(1 - \frac{KT_C}{KT_H}\right) \times 100$$

$$\boxed{\eta_{HE} = \left(1 - \frac{T_C}{T_H}\right) \times 100}$$

Applications :-
1. Design of Steam Turbin engine (Coal Thermal power plants
Nuclear Thermal power plants)

2. Gas Turbin engine
3. ICE (petrol, Diesel, kerosin) engine

Tutorial

1. 10^6 Joules are transferred from a reservoir at 327°C to an engine that operates on the Carnot cycle. The engine rejects heat to a reservoir at 27°C. Determine the thermal efficiency of the cycle and work done by engine

Sol

$$\eta = \left(1 - \frac{T_C}{T_H}\right) \times 100$$

$27 \rightarrow 27 + 273 = 300$

$327 \rightarrow 327 + 273 = 600$

$$\eta = \left(1 - \frac{300}{600}\right) \times 100$$

$$= \eta = \left(1 - \frac{1}{2}\right) \times 100$$

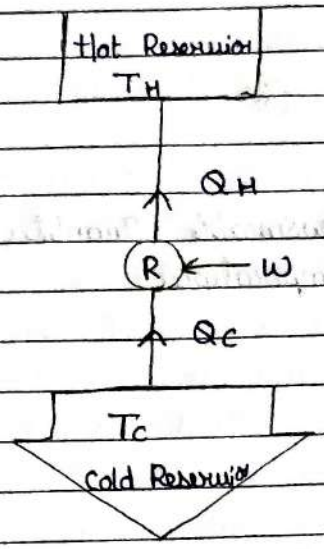
$$= 0.5 \times 100 = 50\%$$

Work done :-

$\eta = W/Q$

$0.5 = W/10^6$

$W = 5 \times 10^5 J$



Refrigerator.

Refrigerator :- It is a device used to maintain cooling effect particularly in house facilities.

Heat energy is removed from cold reservoir (inside refrigerator) and deliver to hot reservoir (to the surroundings). By supplying required work done (or) electricity

The coefficient of performance of refrigerator can be described as:

$$(COP)_R = \frac{\text{Desired O/P}}{\text{Required I/P}}$$

$$(COP)_R = \frac{Q_C}{W} \quad \text{—} (1)$$

Energy Balance :- Energy In = Energy Out

$$Q_C + W = Q_H$$

$$\therefore W = Q_H - Q_C \quad \text{—} (2)$$

Sub ② in ①

$$(COP)_R = \frac{Q_C}{Q_H - Q_C} \quad \text{—③}$$

Q_C & Q_H are unmeasurable Quantities can be estimated through temperatures

$$Q_H \propto T_H$$

$$Q_C \propto T_C$$

$$(COP)_R = \frac{T_C}{T_H - T_C}$$

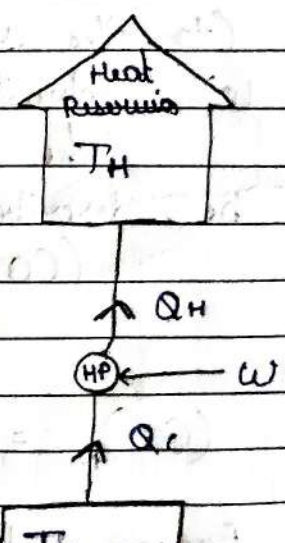
Applications of Refrigerator :-

1. Fridge
2. Freezer
3. Cryogenic preservator
4. Air Conditioner

Heat Pump :-

It is a device used to maintain heating effect.

Heat energy will be removed from cold reservoir and deliver to hot reservoir by ~~by~~ suppling required work done.



The coefficient of performance of heat pump can be :-

$$(COP)_{HP} = \frac{\text{Desired O/P}}{\text{Required I/P}}$$

$$= \frac{\text{Heating effect}}{\text{Required I/P}}$$

$$= \frac{Q_H}{W} \quad \text{—— (1)}$$

$$(COP)_{HP} = \frac{Q_H}{Q_H - Q_C} = \frac{T_H}{T_H - T_C}$$

Tutorial :-

1. A refrigerator system which maintains the refrigerated space at -3°c while operating in a room where temperature is 27°C. Determine the COP of refrigerator system.

Sol $T_c = \cancel{3+2}\ -3+273 = 270°k$

$T_h = 27 + 273 = 300°k$

$$COP = \frac{T_c}{T_h - T_c}$$

$$= 270/(300-270)$$

$$= 9$$

2. prove that $(COP)_{HP} = 1 + (COP)_R$

Sol

Energy balance

$E_{in} = E_{out}$

$$Q_C + W = Q_H$$

Divide with "W"

$$\frac{Q_C}{W} + 1 = \frac{Q_H}{W}$$

$$(COP)_R + 1 = (COP)_{HP}$$

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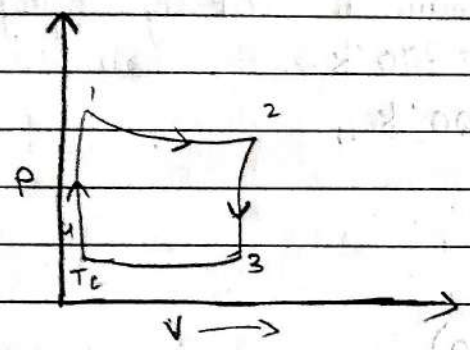
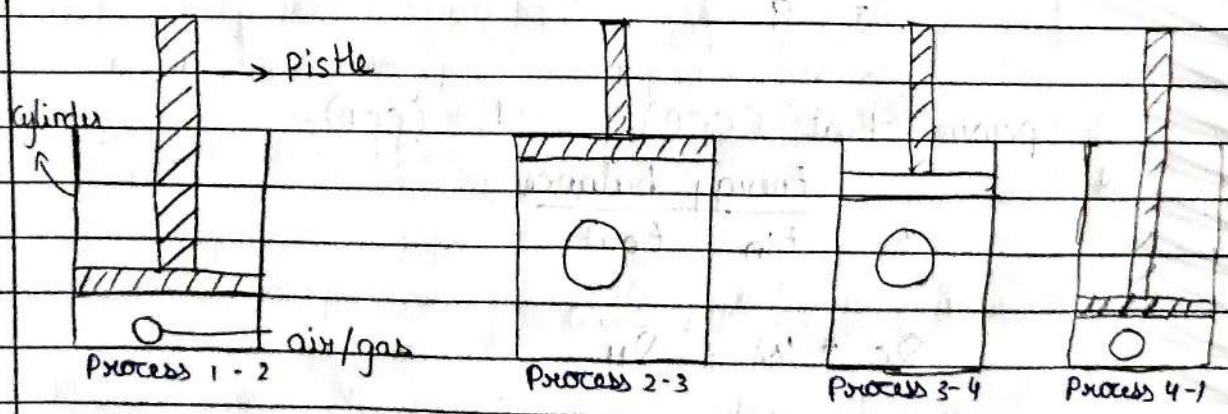
The Carnot Engine Cycle :-
Reversible heat engine

Nicolas Sadi Carnot (1824) introduced the concept of cyclic operations based on 2nd law of thermodynamics.
He designed and developed a classical cycle and renamed after him as a carnot engine.
It is also known as reversible heat engine.
The basic principle behind carnot engine is two isothermal process & two adiabatic process.

It can be explained from following PV diagram.





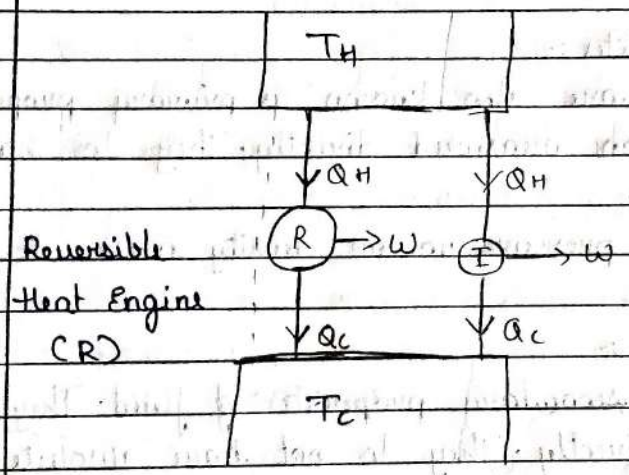
Isothermal expansion (T_H)	Adiabatic expansion (Q_H)	Isothermal compression (T_L)	Adiabatic compression (Q_L)

The Net work done by the engine is estimated as

$$\boxed{W = \Sigma pv}$$

$$W = \underbrace{P_1V_1 + P_2V_2}_{\text{expansion}} + \underbrace{P_3V_3 + P_4V_4}_{\text{compression}}$$

Comparision of Heat Engine & Carnot Engine



Heat Engine (I)

$$\boxed{\eta_I = 0.3 - 0.5}$$

$$\boxed{\eta_Q = 0.5 - 0.8}$$

→ In thermal efficiency of irreversible heat engine is around due to high entropy

→ In thermal efficiency of reversible heat engine is around 0.5 to 0.8.
due to low entropy

—— x ——